

Q. Modular tensor categories, Drinfeld center and anyon models.

4.1. Braided monoidal categories

Def. A braided monoidal cat. consists of

- a monoidal cat. $\mathcal{C}$
- a natural isomorphism call braiding that assigns to every pair of objects $X, Y \in \mathcal{C}$ an iso.

$$b_{X,Y} : X \otimes Y \rightarrow Y \otimes X$$

such that the hexagon eq hold:

$$\begin{array}{ccccc} X \otimes (Y \otimes Z) & \xrightarrow{a_{X,Y,Z}^{-1}} & (X \otimes Y) \otimes Z & \xrightarrow{b_{X,Y} \otimes id_Z} & (Y \otimes X) \otimes Z \\ \downarrow b_{X,Y \otimes Z} & & \downarrow & \searrow & \downarrow a_{Y,X,Z} \\ (Y \otimes Z) \otimes X & \xleftarrow{a_{Y,Z,X}^{-1}} & Y \otimes (Z \otimes X) & \xleftarrow{id_Y \otimes b_{X,Z}} & Y \otimes (X \otimes Z) \end{array}$$

$$\begin{array}{ccccc} (X \otimes Y) \otimes Z & \xrightarrow{a_{X,Y,Z}} & X \otimes (Y \otimes Z) & \xrightarrow{id_X \otimes b_{Y,Z}} & X \otimes (Z \otimes Y) \\ \downarrow b_{X \otimes Y, Z} & & \downarrow & \searrow & \downarrow a_{X,Z,Y} \\ Z \otimes (X \otimes Y) & \xleftarrow{a_{Z,X,Y}} & (Z \otimes X) \otimes Y & \xleftarrow{b_{X,Z} \otimes id_Y} & (X \otimes Z) \otimes Y \end{array}$$

Rem. 1.

$$\begin{array}{ccc} b_{X,Y} = & & b_{X,Y}^{-1} = \\ \begin{array}{c} \text{Diagram of } b_{X,Y} \end{array} & = & \begin{array}{c} \text{Diagram of } b_{X,Y}^{-1} \end{array} \\ b_{Y,X} \circ b_{Y,X}^{-1} = id_{X \otimes Y} = b_{X,Y}^{-1} \circ b_{X,Y} \end{array}$$

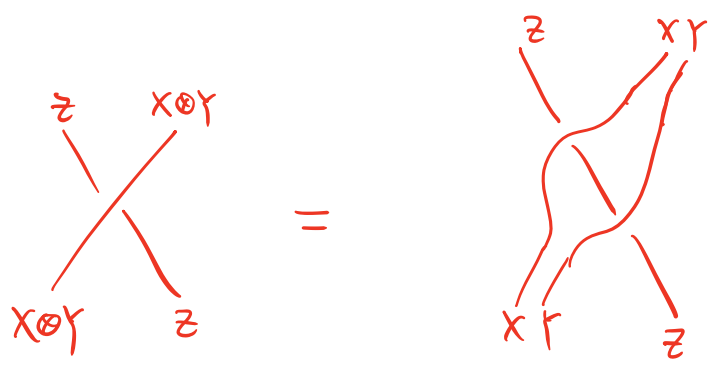
2. hexagon eq. $\Leftrightarrow$

$$\begin{array}{c} \text{Diagram of hexagon eq.} \end{array}$$

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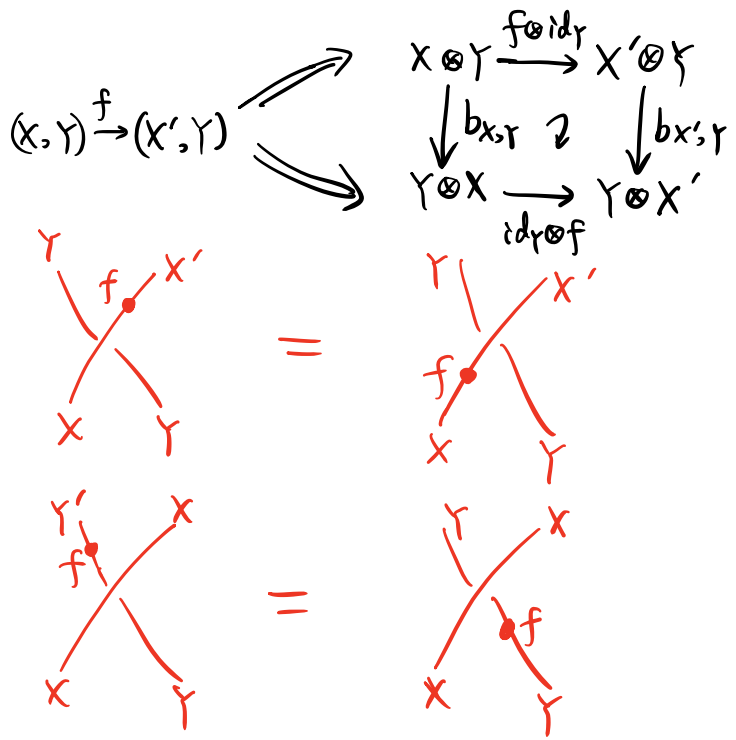
$$b = a^{-1} b a b a^{-1}$$

$$\Leftrightarrow a b a = b a b$$

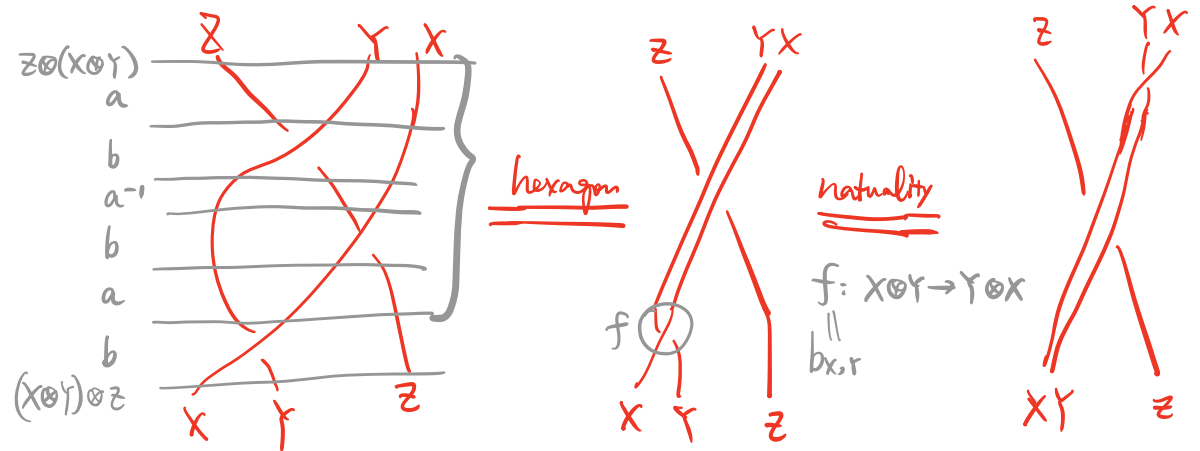


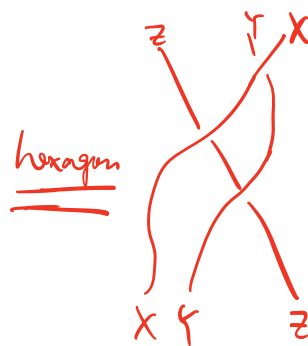
3. naturality of braiding:

$\otimes: \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$

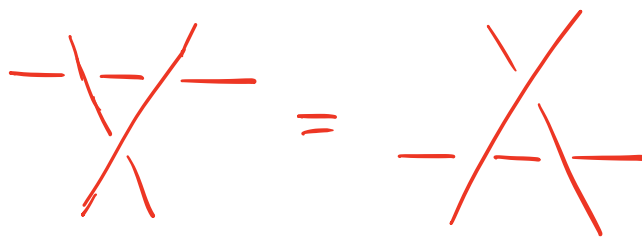


4. naturality + hexagon eq. $\Rightarrow$ Yang-Baxter eq.



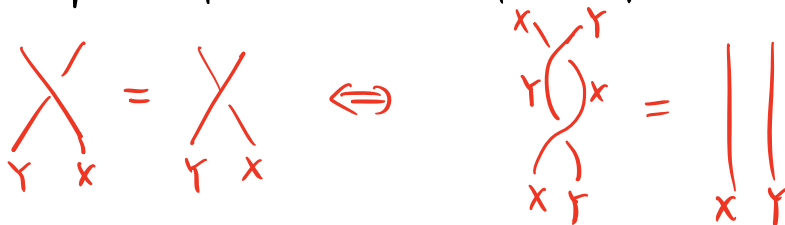


Reidemeister move III for knot $\Leftrightarrow$ Yang-Baxter eq.
(geometry) (algebra)



5. $\mathcal{C}$ is called symmetric monoidal category if

$$b_{x,y}^{-1} = b_{y,x} \Leftrightarrow b_{y,x} \circ b_{x,y} = \text{id}_{x \otimes y}$$



6. String diagram.

Split $a \otimes b = \sum_c N_c^{ab} c$ in the simple obj. basis.



braiding $\rightarrow$ a trivalent graph in 3D.

$$\begin{array}{c} b \quad a \\ \diagdown \quad \diagup \\ a \quad b \\ \diagup \quad \diagdown \\ \mu \\ c \end{array} = \sum_{\mu, \nu} (R_c^{ab})_{\mu, \nu} \begin{array}{c} b \quad a \\ \diagdown \quad \diagup \\ \mu \quad \nu \\ \diagup \quad \diagdown \\ c \end{array}$$

R_c^{ab} is assumed to be unitary.

$$R_{ab} = \begin{array}{c} b \quad a \\ \diagdown \quad \diagup \\ a \quad b \end{array} = \sum_{c, \mu, \nu} \sqrt{\frac{d_c}{d_a d_b}} \begin{array}{c} b \quad a \\ \diagdown \quad \diagup \\ \mu \quad \nu \\ \diagup \quad \diagdown \\ c \end{array} = \sum_{c, \mu, \nu} \sqrt{\frac{d_c}{d_a d_b}} (R_c^{ab})_{\mu, \nu} \begin{array}{c} b \quad a \\ \diagdown \quad \diagup \\ \mu \quad \nu \\ \diagup \quad \diagdown \\ c \end{array}$$

(3D diagram)

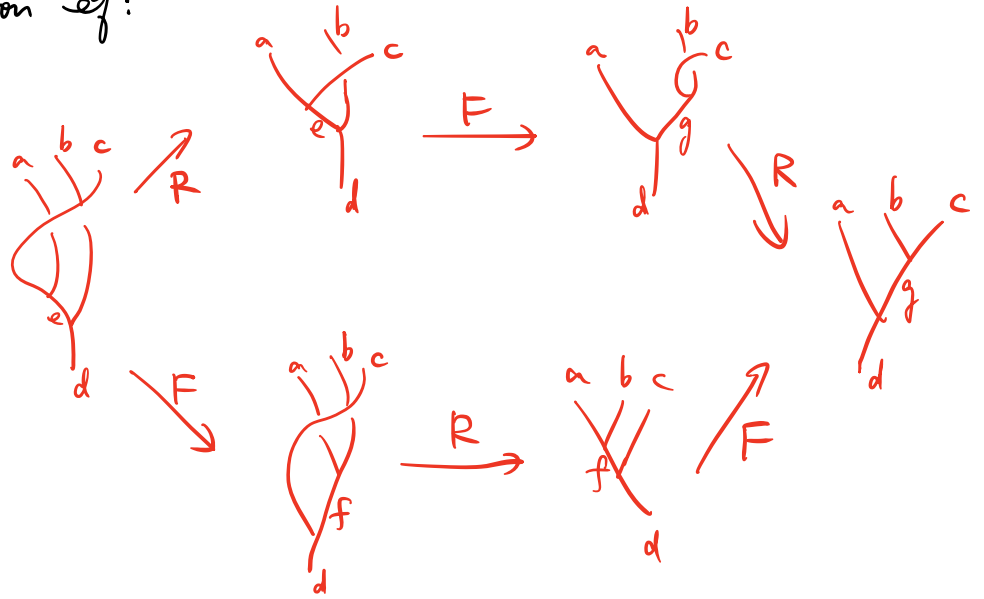


(2D diagram)

naturality :



hexagon eq:



$$\sum (R \cdots) \cdots (F \cdots) \cdots (R \cdots) = \sum F R F$$

$$\forall \text{ 3D trivalent graph } \xrightarrow{R, F} \text{ 2D trivalent graph } \xrightarrow{F} \phi$$

7. modular data.

$$\theta_a = \frac{1}{d_a} \bigcirc_a = \sum_{\mu} \frac{d_c}{d_a} (R_c^{aa})_{\mu\mu}$$

$$\begin{cases} T_{ab} = \theta_a S_{a,b} \\ S_{ab} := \frac{1}{D} a \bigcirc_b = \frac{1}{D} \sum_c N_{ab}^c \frac{\theta_c}{\theta_a \theta_b} d_c \end{cases}$$

 $\mathcal{C}$ is called modular if S, T are non-singular. S, T generate rep of $PSL_2(\mathbb{Z})$.

8. 2+1D anyon models are described by a unitary modular tensor category (UMTC)

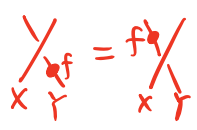
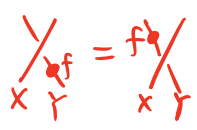
anyon	a	object
anti-particle	a^*	dual object
vacuum	1	identity obj.
fusion of anyons	$a \otimes b$	tensor product
anyon braiding	$R_{a,b}$	braiding
anyon worldline in 2+1D		string diagram in 3D
transition amplitude		linear map: $a_1 \otimes a_2 \otimes \dots \rightarrow b_1 \otimes b_2 \otimes \dots$
partition function	closed 3D diagram	linear map: $1 \rightarrow 1$

4.2. Drinfeld center construction.

Let $\mathcal{C}$ be a monoidal category, a half braiding β_X for $X \in \mathcal{C}$ is a family $\{\beta_X(Y) \in \text{Hom}_{\mathcal{C}}(X \otimes Y, Y \otimes X) \mid Y \in \mathcal{C}\}$ of iso,

natural w.r.t. Y , satisfying $\beta_X(1) = \text{id}_X$ and $\beta_X(Y \otimes Z) = [\text{id}_Y \otimes \beta_X(Z)] \circ [\beta_X(Y) \otimes \text{id}_Z]$

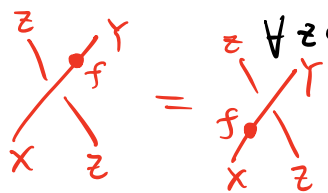
Diagrammatic representations of the conditions:

- $\beta_X(1) = \text{id}_X$: 
- Naturality: 
- Half braiding property: 

The Drinfeld center $\mathcal{Z}(\mathcal{C})$ of $\mathcal{C}$ has obj. (X, β_X) , where $X \in \mathcal{C}$ and β_X is a half braiding for X .

The morphisms are

$$\text{Hom}_{\mathcal{Z}(\mathcal{C})}((X, \beta_X), (Y, \beta_Y)) := \{f \in \text{Hom}_{\mathcal{C}}(X, Y) \mid [\text{id}_Y \otimes f] \circ \beta_X(Z) = \beta_Y(Z) \circ [f \otimes \text{id}_X], \forall Z \in \mathcal{C}\}$$

Diagrammatic representation of the condition: 

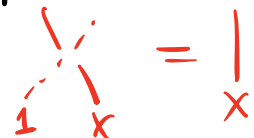
The tensor product in $\mathcal{Z}(\mathcal{C})$ is given by

$$(X, \beta_X) \otimes (Y, \beta_Y) = (X \otimes Y, \beta_{X \otimes Y})$$

where $\beta_{X \otimes Y}(Z) := [\beta_X(Z) \otimes \text{id}_Y] \circ [\text{id}_X \otimes \beta_Y(Z)]$

Diagrammatic representation: 

The tensor unit is $(1, \beta_1)$ where $\beta_1(X) := \text{id}_X$.

Diagrammatic representation: 

The composition and tensor product of morphisms in $\mathcal{Z}(\mathcal{C})$ are inherited from $\mathcal{C}$.

The braiding in $\mathcal{Z}(\mathcal{C})$ is given by

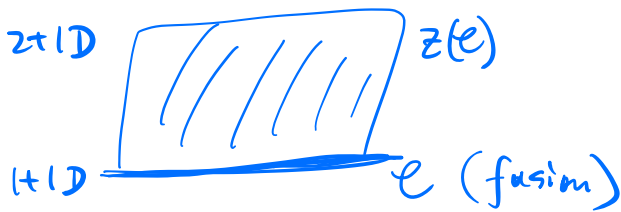
$$b_{(X, \beta_X), (Y, \beta_Y)} := \beta_X(Y)$$

$$\begin{array}{c} \beta_x(r) \\ \diagup \quad \diagdown \\ (x, \beta_x) \quad (y, \beta_y) \end{array}$$

$\Rightarrow \mathcal{Z}(\mathcal{C})$ is a braided monoidal cat.

- Rem. 1. $\mathcal{C}$ is monoidal $\xrightarrow{\mathcal{Z}(\cdot)}$ $\mathcal{Z}(\mathcal{C})$ is braided monoidal
2. If $\mathcal{C}$ is fusion, then $\mathcal{Z}(\mathcal{C})$ is modular,
and $\dim \mathcal{Z}(\mathcal{C}) = (\dim \mathcal{C})^2$, where $\dim \mathcal{C} := \sum_n d_n^2$
3. If $\mathcal{C}$ is modular tensor cat, then
 $\mathcal{Z}(\mathcal{C}) = \mathcal{C} \boxtimes \mathcal{C}^{\text{op}}$.

$\downarrow$
same as $\mathcal{C}$ with inverse braiding



4. String diagram in simple object basis.

$$\begin{array}{c} \diagdown \quad \diagup \\ (a, \beta_a) \quad b \end{array} = \beta_a(b) \begin{array}{c} b \quad a \\ \diagdown \quad \diagup \\ c \\ \diagup \quad \diagdown \\ a \quad b \end{array}$$

$$(N_c^{ab} = 1) \\ a \otimes b = c = ab \text{ is simple.}$$

naturality:

$$\begin{array}{c} a \quad b \quad c \\ \diagdown \quad \diagup \\ c \quad a \otimes b \\ \parallel \\ c \quad ab \end{array} = \begin{array}{c} a \quad b \quad c \\ \diagdown \quad \diagup \\ c \quad ab \end{array}$$

$$\text{LHS} = \beta_c(a \otimes b)$$

$$\text{RHS} = \beta_c(a) \beta_c(b)$$

$$\begin{array}{c} a \\ \diagup \quad \diagdown \\ c \quad b^* \end{array} = (?) \cdot \begin{array}{c} a \quad b \\ \diagup \quad \diagdown \\ c \quad b^* \end{array}$$

$$\text{Hom}(c \otimes b^*, a) \cong \text{Hom}(c, a \otimes b)$$

$$\begin{array}{c} a \\ \diagup \quad \diagdown \\ c \quad b^* \\ \parallel \\ a \end{array} = (?) \cdot \begin{array}{c} a \quad b \\ \diagup \quad \diagdown \\ c \quad b^* \\ \parallel \\ F_a^{a,b,b^*} \end{array}$$

$$\parallel$$

$$\begin{array}{c} \sqrt{\frac{d_c d_b}{d_a}} \\ \parallel \\ a \end{array}$$

$$\parallel$$

$$\begin{array}{c} F_a^{a,b,b^*} \\ \parallel \\ F_a^{a,b,b^*} d_b \\ \parallel \\ a \end{array}$$

$$(?) = \sqrt{\frac{d_a}{d_b d_c}} d_b F_a^{a,b,b^*} = \sqrt{\frac{d_a d_b}{d_c}} F_a^{a,b,b^*}$$

↓ assume

$$1$$

$$\text{LHS} = \beta_c(a \otimes b)$$

$$\parallel$$

Naturality

$$\text{RHS} = \beta_c(a) \beta_c(b)$$

$$= \beta_c(a) \beta_c(b) \left(F_{abc,b^*}^{a,b,c} \right)^{-1}$$

$$= \beta_c(a) \beta_c(b) \left(F_{abc,b^*}^{a,b,c} \right)^{-1} \left(F_{abc}^{a,b,c} \right)^{-1}$$

$$= \beta_c(a) \beta_c(b) \left(F_{abc,b^*}^{a,b,c} \right)^{-1} \left(F_{abc}^{a,b,c} \right)^{-1}$$

$$F_{abc,b^*,a^*}^{abc,b^*,a^*}$$

$$F_{abc,b^*,a^*}^{abc,b^*,a^*}$$

$$\begin{array}{c} ab \\ \circ \\ a \quad b \\ \parallel \\ ab \end{array} = \sqrt{\frac{d_a d_b}{d_{a \otimes b}}} \parallel = \parallel$$

$$d_{a \otimes b} = \text{area of circle} = d_a \cdot d_b$$

$$= \beta_c(a) \beta_c(b) (F_{abc, b^*}^{a, bc, b^*})^{-1} (F_{abc}^{a, b, c})^{-1} F_{abc, b^*, a^*}^{abc, b^*, a^*}$$



$$\Rightarrow \beta_c(a \otimes b) = \beta_c(a) \beta_c(b) (F_{abc, b^*}^{a, bc, b^*})^{-1} (F_{abc}^{a, b, c})^{-1} F_{abc, b^*, a^*}^{abc, b^*, a^*}$$



$\beta_c(-)$ half braiding of c .

5. Example of $\mathcal{Z}(\text{Vec}_{\mathbb{Z}_2})$: toric code. $\mathbb{Z}_2$ gauge theory.
 $\nu_3 \rightarrow \text{nontrivial} \in H^3(\mathbb{Z}_2, \mathbb{U}(1))$ double semion model

$$\mathcal{C} = \text{Vec}_{\mathbb{Z}_2} \quad \text{obj.} = \{1, a\}, \quad a \otimes a = 1$$

$$\beta_1(1), \beta_1(a); \quad \beta_a(1), \beta_a(a)$$

$$\begin{cases} \beta_1(a): \\ \beta_a(1): \end{cases} \quad \begin{array}{c} \text{Diagram 1: } a \text{ and } 1 \text{ crossing, } a \text{ on top, } 1 \text{ on bottom.} \\ \text{Diagram 2: } a \text{ and } 1 \text{ crossing, } 1 \text{ on top, } a \text{ on bottom.} \end{array} \neq$$

$$\textcircled{1} \beta_1(a): \quad \begin{array}{c} \text{Diagram 1: } a \text{ and } 1 \text{ crossing, } a \text{ on top, } 1 \text{ on bottom.} \\ \text{Diagram 2: } a \text{ and } 1 \text{ crossing, } 1 \text{ on top, } a \text{ on bottom.} \end{array} = \beta_1(a) \cdot \begin{array}{c} \text{Diagram 3: } a \text{ and } 1 \text{ crossing, } a \text{ on top, } 1 \text{ on bottom.} \end{array}$$

$$\begin{array}{c} \text{Diagram 4: } a \text{ and } 1 \text{ crossing, } a \text{ on top, } 1 \text{ on bottom.} \\ \text{Diagram 5: } a \text{ and } 1 \text{ crossing, } 1 \text{ on top, } a \text{ on bottom.} \end{array} = \begin{array}{c} \text{Diagram 6: } a \text{ and } 1 \text{ crossing, } a \text{ on top, } 1 \text{ on bottom.} \\ \text{Diagram 7: } a \text{ and } 1 \text{ crossing, } 1 \text{ on top, } a \text{ on bottom.} \end{array}$$

$$\left. \begin{array}{l} \text{|| naturality} \\ \text{Diagram 8: } a \text{ and } 1 \text{ crossing, } a \text{ on top, } 1 \text{ on bottom.} = [\beta_1(a)]^2 \text{ Diagram 9: } a \text{ and } 1 \text{ crossing, } a \text{ on top, } 1 \text{ on bottom.} = [\beta_1(a)]^2 \text{ Diagram 10: } a \text{ and } 1 \text{ crossing, } 1 \text{ on top, } a \text{ on bottom.} \end{array} \right\} \Rightarrow \beta_1(a) = \pm 1$$

$$\textcircled{2} \beta_a(a): \quad \begin{array}{c} \text{Diagram 11: } a \text{ and } a \text{ crossing, } a \text{ on top, } a \text{ on bottom.} \\ \text{Diagram 12: } a \text{ and } a \text{ crossing, } a \text{ on top, } a \text{ on bottom.} \end{array} = \beta_a(a) \cdot \begin{array}{c} \text{Diagram 13: } a \text{ and } a \text{ crossing, } a \text{ on top, } a \text{ on bottom.} \end{array}$$

$$\begin{array}{c} \text{Diagram 14: } a \text{ and } a \text{ crossing, } a \text{ on top, } a \text{ on bottom.} \\ \text{Diagram 15: } a \text{ and } a \text{ crossing, } a \text{ on top, } a \text{ on bottom.} \end{array} = \begin{array}{c} \text{Diagram 16: } a \text{ and } a \text{ crossing, } a \text{ on top, } a \text{ on bottom.} \\ \text{Diagram 17: } a \text{ and } a \text{ crossing, } a \text{ on top, } a \text{ on bottom.} \end{array} = F_a^{aaa} \cdot \begin{array}{c} \text{Diagram 18: } a \text{ and } a \text{ crossing, } a \text{ on top, } a \text{ on bottom.} \end{array}$$

$\Rightarrow \beta_a(a) = \pm 1$ for $\nu_3 \in H^3(\mathbb{Z}_2, \mathbb{U}(1))$ (semion)
 $\beta_a(a) = \pm i$ for $\nu_3 \in H^3$

$$\left. \begin{array}{l} \text{|| naturality} \\ \text{Diagram 19: } a \text{ and } a \text{ crossing, } a \text{ on top, } a \text{ on bottom.} = [\beta_a(a)]^2 \text{ Diagram 20: } a \text{ and } a \text{ crossing, } a \text{ on top, } a \text{ on bottom.} = [\beta_a(a)]^2 \text{ Diagram 21: } a \text{ and } a \text{ crossing, } a \text{ on top, } a \text{ on bottom.} \end{array} \right\}$$

$\Rightarrow$ There are 4 objects (X, β_X) in $\mathcal{Z}(\text{Vec}_{\mathbb{Z}_2})$
 $\underbrace{(1, \beta_1(1)=1, \beta_1(a)=+1)}_{1 \in \mathcal{Z}(\text{Vec}_{\mathbb{Z}_2})}, \underbrace{(1, \beta_1(1)=1, \beta_1(a)=-1)}_e, \underbrace{(a, \beta_a(1)=1, \beta_a(a)=+1)}_m, \underbrace{(a, \beta_a(1)=1, \beta_a(a)=-1)}_f$

$\Rightarrow$ Toric code = $\mathbb{Z}_2$ gauge theory has 4 excitations.

Statistics :



$$b_{e,m} = b_{\underbrace{(1, \beta_1(a)=-1)}_e, \underbrace{(a, \beta_a(a)=+1)}_m} = \beta_1(a) = -1$$

$b(x, \beta_x), (y, \beta_y) := \beta_x(y)$

$$b_{m,e} = b_{(a, \beta_a(a)=+1), (1, \beta_1(a)=-1)} = \beta_a(1) = 1$$

$$\begin{array}{c} \diagup \diagdown \\ e \quad m \end{array} = b_{me} \circ b_{em} \begin{array}{c} || \\ e \quad m \end{array} = \beta_a(1) \cdot \beta_1(a) \begin{array}{c} || \\ e \quad m \end{array} = - \begin{array}{c} || \\ e \quad m \end{array}$$

4.3. Excitation in Levin-Wen model of fusion cat. $\mathcal{C}$.

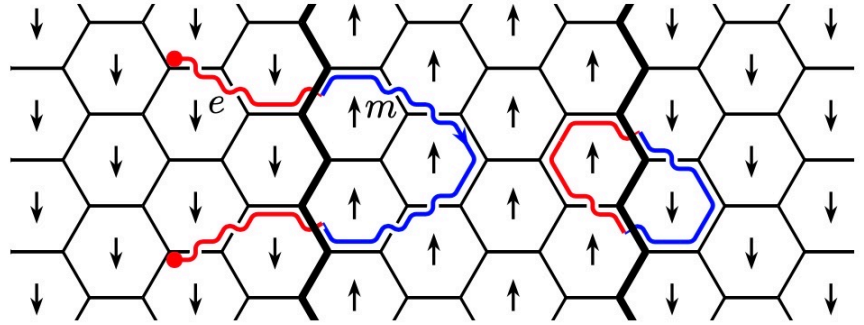
↳ described by $\mathcal{Z}(\mathcal{C})$

(1) diagram for ribbon operators.

$$|\blacksquare \circ^\alpha\rangle = \sum_i n_{\alpha,i} |\blacksquare \circ^i\rangle$$

$$|\begin{array}{c} \alpha \\ \diagdown \quad \diagup \\ i \end{array}\rangle = \sum_{jst} (\Omega_{\alpha,sti}^j)_{\sigma\tau} \left| \begin{array}{ccc} i & j & s \\ t & & i \end{array} \right\rangle$$

$$|\begin{array}{c} \diagdown \quad \diagup \\ \alpha \\ i \end{array}\rangle = \sum_{jst} (\bar{\Omega}_{\alpha,sti}^j)_{\sigma\tau} \left| \begin{array}{ccc} t & j & i \\ i & & s \end{array} \right\rangle$$



ribbon op. are labelled by obj. in $\mathcal{C}$.

$$\begin{array}{c} \diagdown \quad \diagup \\ a \quad b \end{array} = \sum R \begin{array}{c} \diagdown \quad \diagup \end{array}$$

$$[\text{ribbon op.}, \begin{smallmatrix} A_s \\ B_p \end{smallmatrix}] = 0 \text{ if } s, p \notin \partial \text{ ribbon.}$$

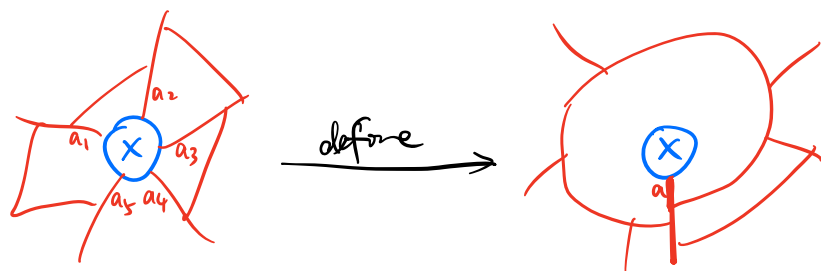
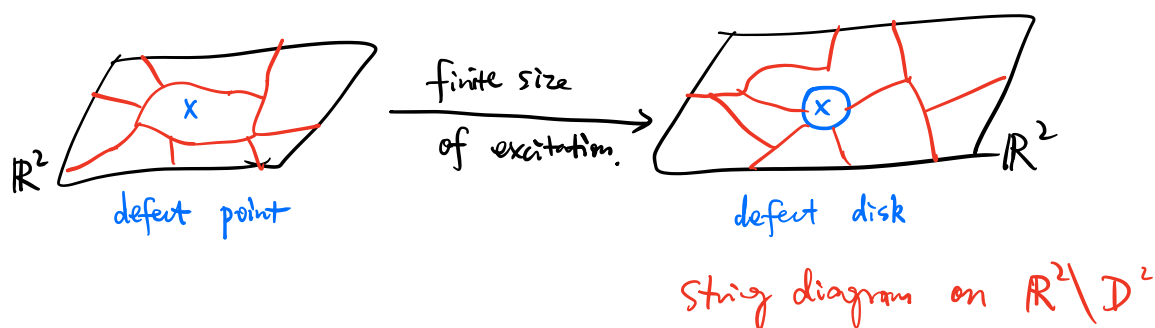
$$\begin{array}{c} \diagdown \quad \diagup \\ \diagup \end{array} = \begin{array}{c} \diagdown \quad \diagup \\ \diagdown \end{array} \quad \text{naturality}$$

...

⇒ ribbon op. are labelled by $\mathcal{Z}(\mathcal{C})$.

(2) Tube algebra approach

excitation = defect of the wavefunction or Hamiltonian.

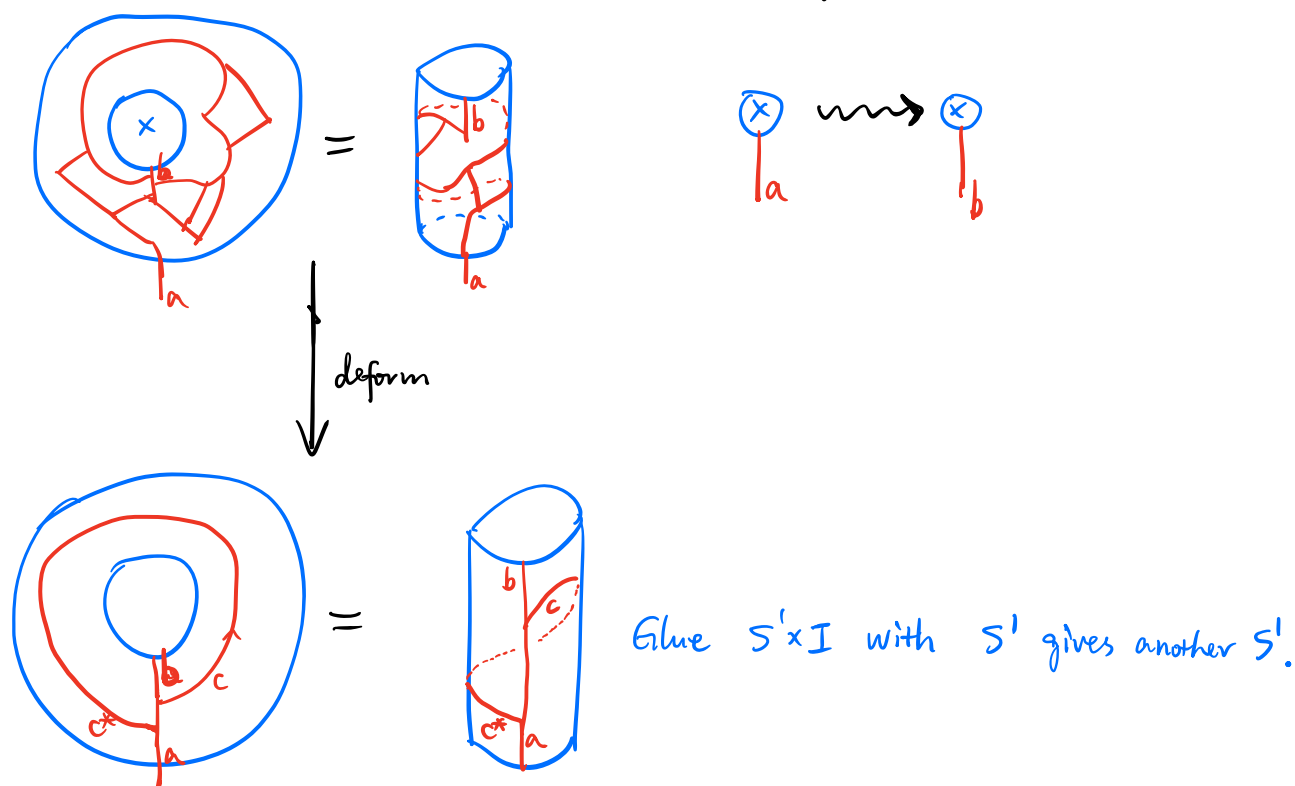


$\Rightarrow$ excitations are labelled by obj. $a \in \mathcal{C}$.

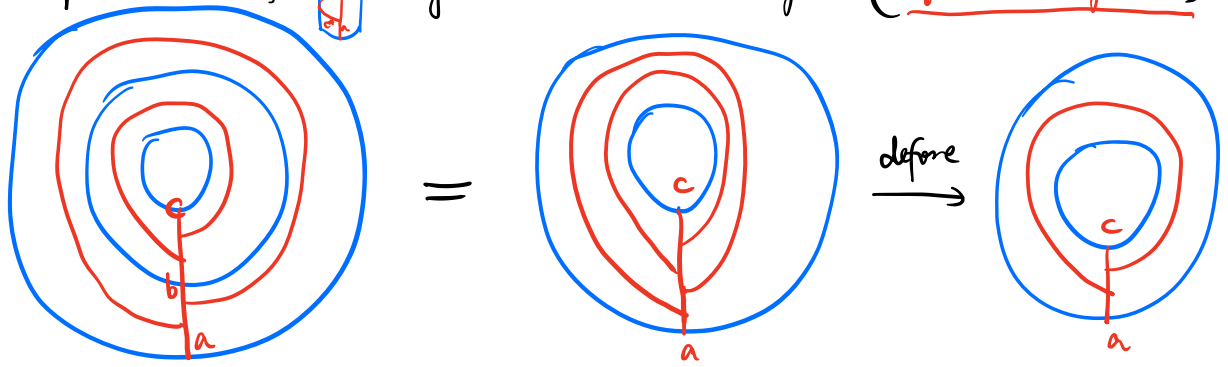
Q: What is the morphism between excitations?

$\leftrightarrow$ How to change one excitation into another?

A: Glue a cylinder $S^1 \times I$ with string diagram.



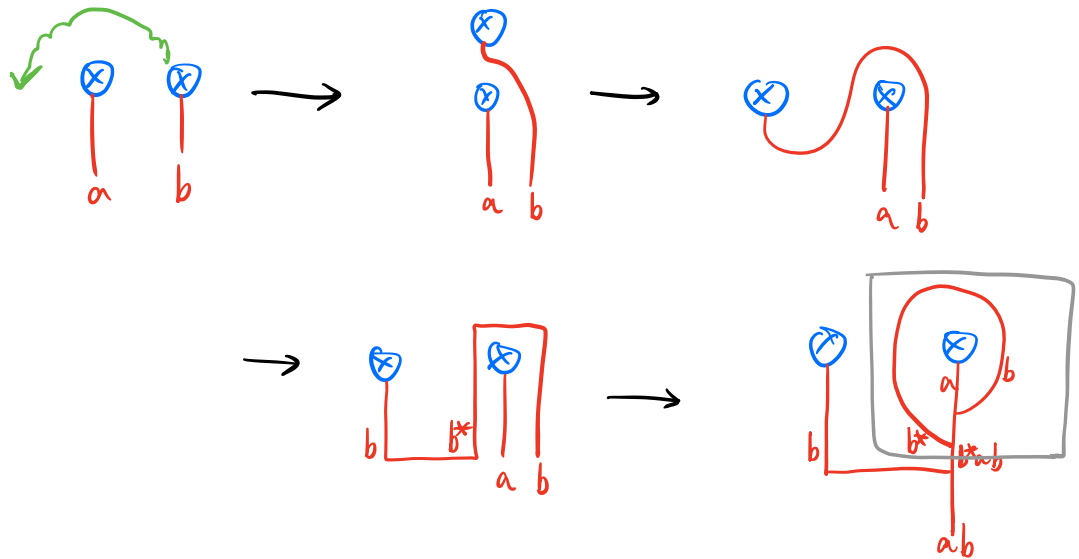
Composition of  gives us an algebra (Tube algebra)



$\Rightarrow$ Excitations can be acted by tube algebra

$\Rightarrow$ Excitations are reps of tube algebra $\left. \begin{array}{l} \text{cont. of rep. of tube algebra} = \mathbb{Z}(c) \\ \text{?} \end{array} \right\} \Rightarrow \text{excitations} \in \mathbb{Z}(c)$

Braiding of excitations = defects:



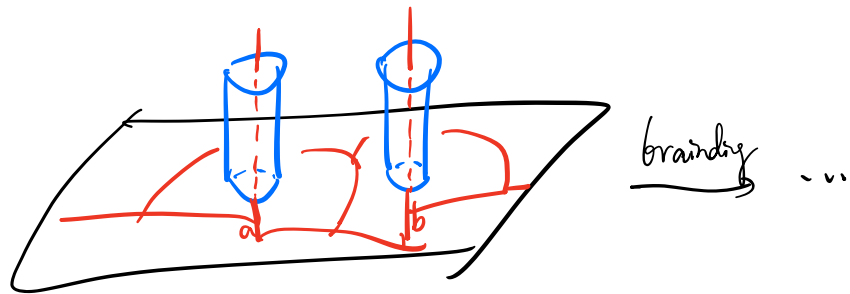
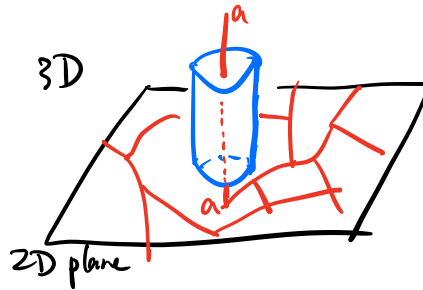
$$|a, b\rangle \xrightarrow{\text{braiding}} |b, \text{tube algebra } a\rangle$$

$$\rightsquigarrow |(a, \beta_a), (b, \beta_b)\rangle \xrightarrow{\text{braiding}} |(b, \beta_b), \beta_b(-)(a, \beta_a)\rangle$$

$$\text{tube algebra } a = \beta_b(a) \text{ tube algebra } b$$



Excitations can be understood as



$\Rightarrow$ Excitations should be labelled by $(a, \beta_a) \in Z(\mathcal{C})$.